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Policy Uncertainty in the Market for Coal Electricity: The Case of Air Toxics Standards

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1 Big picture

- **Question.** How much policy uncertainty did coal generators face about the eventual enforcement of the Mercury and Air Toxics Standards (MATS), and how would outcomes change if that uncertainty were resolved earlier?
- **Setting.** The paper studies merchant coal-fired electricity generators (independent power producers, or IPPs) in the US Eastern Interconnection from 2006–2017, focusing on their decisions to adopt pollution-abatement technology or exit.
- **Core challenge.** Policy uncertainty matters for irreversible decisions, but in this setting firms are also strategic oligopolists: one generator’s exit or adoption changes rivals’ future profits.
- **Main methodological result.** The paper estimates a dynamic oligopoly model that recovers firms’ subjective beliefs about future MATS enforcement and introduces *approximate belief oligopoly equilibrium* (ABOE) to make equilibrium computation feasible with many generators.
- **Main empirical result.** The perceived probability that MATS would be enforced in 2016 was essentially 1 in 2012, fell to 0.738 in 2013 and 0.428 in 2014, and rose back to essentially 1 in 2015.
- **Main counterfactual result.** Resolving uncertainty immediately in 2012 would raise generators’ expected profits by \$1.39 billion, but it would also increase expected SO₂ pollution by about 52.5 million pounds, with damages valued at \$652 million to \$1.776 billion.
- **Economic takeaway.** Earlier uncertainty resolution is not automatically environmentally beneficial. When many firms are close to the exit margin, learning early that regulation may not bind can keep more polluting firms alive.

2 Data and measurement (key objects)

2.1 Policy setting and observed data

- **Policy background.** The EPA’s first major mercury rule, CAMR, was finalized in 2005 and vacated in 2008. After further court decisions, the EPA finalized MATS in 2012, with effective compliance largely by 2016.
- **Source of uncertainty.** Uncertainty came from both court challenges and changes in executive leadership.
- **Main operational data.** The core dataset is the EPA *Continuous Emissions Monitoring System* (CEMS), which gives hourly generator-level heat input, electricity generation, and emissions.
- **Additional data.**
 - EPA annual generator characteristics,
 - EIA Form 923 for facility and operator information,
 - EIA Form 423 for fuel prices,
 - hourly wholesale electricity prices from major ISOs/RTOs,

- PUDL/FERC 714 load data, and
- PRISM weather data.
- **Sample.** The paper studies coal IPPs in the Eastern Interconnection from 2006 to 2017.
- **Key measurement choice.** Because actual MATS-compliance technologies are hard to observe directly, the paper uses SO₂ emissions behavior as a proxy for abatement-technology adoption and also as the main pollution outcome.

2.2 Markets, generators, and regulatory regimes

- **Market definition.** A market is a US state.
- **Generator definition.** The paper works at the EPA “unit” level, essentially emissions release points. A generator is coal if its primary fuel includes the word “coal.”
- **Firm focus.** The paper focuses on merchant generators / IPPs because they face market incentives rather than cost-of-service regulation.
- **Regulatory comparison.**
 - Some states adopted their own air-toxics standards before MATS.
 - These state standards were generally much less uncertain once implemented.
 - But they were often weaker than MATS, so the model allows MATS compliance costs to differ from state-standard compliance costs.
- **Outside competition.** Natural gas, renewables, utility-owned coal, and other generation sources affect profits, but their dynamic entry, exit, adoption, and production decisions are treated as exogenous.

2.3 Key descriptive objects

- **Sample size.** 310 IPP coal generators and 2,791 generator-year observations.
- **Regime split.** 93 generators are subject to state air-toxics standards, while 217 are subject to MATS.
- **Technology and cost comparison (Table 1).**
 - Mean capacity: 279.69 MW under state standards vs. 246.01 MW under MATS.
 - Mean heat rate: 10.51 vs. 11.99.
 - Mean coal fuel price: \$1.77 vs. \$2.20 per MMBtu.
 - Mean marginal fuel cost: \$18.48 vs. \$27.29 per MWh.
- **Compliance pattern (Table 2).**
 - State-standard generators: 30 adoptions and 5 exits.
 - MATS generators: 14 adoptions and 69 exits.
- **Interpretation.** State-standard compliance mainly happened through adoption, while MATS compliance mainly happened through exit.

- **Hourly operating facts (Table 3).**

- At maximum generation: 45% of hours.
- At minimum generation: 24%.
- Off: 31%.
- Mean generation: 177.02 MWh.
- Mean price: \$39.38/MWh.

2.4 Unobservables and timing

- **Time-invariant generator traits.** Heat rate, capacity, location, and fixed coal-price environment.
- **Time-varying generator traits.** Whether the unit has adopted abatement technology, the belief year, years to possible enforcement, and private shocks to adopt / continue / exit.
- **Private shocks.** Annual adoption / continue / exit shocks are i.i.d. type I extreme value. Hourly operating choices also have idiosyncratic ramping shocks.
- **Timing within a year.**
 1. The policy environment updates and firms revise beliefs.
 2. Generators choose adopt / continue / exit.
 3. They earn annual profits from hourly wholesale-market competition.
 4. Decisions are realized; if this is the enforcement year and MATS is enforced, nonadopters are forced to exit.
- **Limitation.** Each generator is modeled as an independent optimizer; the paper does not explicitly model coordinated optimization across multiple units owned by the same firm.

3 Models and empirical specifications (all equations + interpretation)

3.1 Simple intuition: ex post vs. ex ante uncertainty resolution

Let B denote expected discounted profit absent regulation, C the compliance cost, and p the probability that regulation is enforced.

With **ex post** uncertainty resolution, the generator decides before learning the policy outcome, so it compares

$$B - pC$$

to zero.

Example 1. Suppose

$$B = 150, \quad C = 200, \quad p = 0.5.$$

Then

$$150 - 0.5 \cdot 200 = 50 > 0,$$

so the plant stays.

With **ex ante** uncertainty resolution, the plant learns first whether the rule will be enforced.

- If enforced, it exits.
- If not enforced, it stays.

So exit probability becomes 0.5. Here, earlier resolution *increases* exit.

Example 2. Suppose instead

$$B = 50, \quad C = 200, \quad p = 0.5.$$

Then

$$50 - 0.5 \cdot 200 = -50 < 0,$$

so with ex post resolution, the plant exits for sure.

But with ex ante resolution:

- it exits only if enforcement occurs,
- so exit probability is again 0.5.

Here, earlier resolution *decreases* exit.

Interpretation. The effect of resolving uncertainty earlier depends on where firms sit relative to the exit margin. That is the central intuition behind the empirical results.

3.2 Dynamic oligopoly model and ABOE

- The model is an infinite-horizon dynamic oligopoly with discount factor $\beta = 0.95$.
- Each year, generators decide whether to:
 - adopt abatement technology,
 - continue operating without adopting, or
 - exit.
- The key aggregate market state is

$$\Omega_t = (\text{coal share}, \text{gas-to-coal fuel price ratio}, \text{adoption share}).$$

- Here:
 - **coal share** = coal capacity divided by the 95th percentile of hourly load,
 - **fuel price ratio** = gas price / coal price,
 - **adoption share** = share of IPP coal capacity that has adopted.
- Generator-specific states include heat rate, capacity, whether the unit has adopted, the belief year t , years to potential enforcement $\tilde{t}$, and private shocks.
- The paper's new equilibrium concept is **approximate belief oligopoly equilibrium (ABOE)**.
 - Firms do *not* track the full state of every rival.

- Instead, they track a small set of aggregate market states.
- They still understand that *their own actions affect future market states*.
- Their beliefs about state transitions are approximated by simple conditional moments that are made consistent with simulated equilibrium play.

Interpretation. ABOE is the paper's way of keeping the model strategic, while still making equilibrium computation feasible in a market with many generators.

3.3 Generator Bellman problems

For a generator that has *not* yet adopted and is *one year* from possible enforcement ($\tilde{t} = 1$), the value function is

$$V_j(\Omega, 0, t, 1, \varepsilon_j) = \max\{\text{continue, adopt, exit}\}.$$

The three choice-specific values are:

$$\text{continue} = \Pi_j(\Omega) + P_t X$$

$$+ (1 - P_t)\beta E[V_j(\Omega', 0, t, 0) \mid \Omega, \text{No Standard}] + \varepsilon_j^c,$$

$$\text{adopt} = \Pi_j(\Omega) - A$$

$$+ \beta \left(P_t E[V_j(\Omega', 1, t, 0) \mid \Omega, \text{Standard}] + (1 - P_t) E[V_j(\Omega', 1, t, 0) \mid \Omega, \text{No Standard}] \right) + \varepsilon_j^a,$$

$$\text{exit} = \Pi_j(\Omega) + X + \varepsilon_j^x.$$

Here:

- A = adoption cost,
- X = scrap value from exit (estimated to be negative in practice, so exiting is costly),
- P_t = subjective probability that MATS is enforced,
- $\Pi_j(\Omega)$ = annual operating profit.

For a nonadopter with *more than one year* to enforcement ($\tilde{t} > 1$):

$$V_j(\Omega, 0, t, \tilde{t}, \varepsilon_j) = \max \left\{ \begin{aligned} &\Pi_j(\Omega) + \beta E[V_j(\Omega', 0, t, \tilde{t} - 1)] + \varepsilon_j^c, \\ &\Pi_j(\Omega) - A + \beta E[V_j(\Omega', 1, t, \tilde{t} - 1)] + \varepsilon_j^a, \\ &\Pi_j(\Omega) + X + \varepsilon_j^x \end{aligned} \right\}.$$

Interpretation.

- At $\tilde{t} = 1$, beliefs about enforcement enter directly.
- Adoption does not directly raise operating profits; it matters because it avoids forced exit and can strategically signal staying power to rivals.
- If a generator already adopted, it chooses only between continue and exit.

3.4 Annual profit model

Generators earn annual profits by competing in hourly electricity markets. A reduced representation is

$$\Pi_{jt} = \sum_h q_{jh}(p_h - \text{heat}_j \cdot f^C - om) - \sum_h r_{j,\tilde{q},q} \mathbf{1}\{\tilde{q}_{jh} < q_{jh}\} + \text{ramping-shock term},$$

where

- q_{jh} = generation in hour h ,
- p_h = wholesale electricity price,
- $\text{heat}_j \cdot f^C$ = fuel cost per MWh,
- om = O&M cost per MWh,
- $r_{j,\tilde{q},q}$ = ramping / start-up cost from moving from previous output $\tilde{q}$ to current output q .

Each hour, a coal unit chooses among:

- full capacity K_j ,
- minimum generation $L_j K_j$,
- zero generation.

Interpretation.

- Profit depends not only on price-cost margins but also on cycling and start-up costs.
- This matters a great deal after fracking lowered natural gas prices, because coal plants cycled more often.
- The paper explicitly models this instead of treating annual profits as a black box.

3.5 Profit and pollution surfaces

The dynamic model needs profits and pollution at many possible states, including counterfactual states not directly observed in the data. So the authors estimate flexible reduced-form “surfaces.”

Profit surface:

$$\hat{\Pi}_{jt} = g(\text{coal share, fuel ratio, heat rate}_j, K_j, \text{interactions, US-state FE}).$$

Pollution surface:

$$\log SO2_{jt} = h(\text{compliant}_{jt}, \text{coal share, fuel ratio, heat rate}_j, K_j, \text{US-state FE}).$$

Key details:

- both are estimated by weighted OLS,
- each state-year gets equal weight,
- the pollution regression includes a compliance indicator and uses only nonzero-pollution observations.

Interpretation.

- The profit surface maps economic states into expected annual profits.
- The pollution surface maps states into expected SO₂ emissions.
- These surfaces let the paper run policy counterfactuals without solving a full hourly dispatch game for every hypothetical state.

3.6 Estimation and counterfactual design

The estimation proceeds in stages:

1. Estimate ramping and O&M costs from hourly behavior.
2. Construct annual profits for observed generator-years.
3. Estimate the profit and pollution surfaces.
4. Estimate long-run dynamic parameters:
 - P_t : perceived enforcement probabilities,
 - A : adoption cost,
 - X : exit scrap value,
 - $1/\sigma$: scale of annual action shocks.

The long-run parameters are estimated by nested fixed-point simulated maximum likelihood:

- for each candidate parameter vector, solve the ABOE,
- compute the probability of the observed action in each generator-year,
- maximize the simulated likelihood.

Why not a CCP approach?

- Because subjective enforcement beliefs change over time, later data do not directly reveal the same beliefs firms had earlier.

Main counterfactuals:

- the estimated model,
- same mean enforcement probability (0.7913) in every year, resolved ex post,
- same mean probability, but resolved immediately in 2012 (ex ante),
- no exit cost.

Interpretation. The comparison between the two “same mean probability” cases isolates the effect of changing only the *timing of uncertainty resolution*.

4 Identification and instruments (what makes the estimates credible)

4.1 What is hard to identify

- A generator may delay exit or adoption because:
 - it thinks MATS will not be enforced,
 - adoption is expensive,
 - exit is expensive,
 - or profits are temporarily high.
- So perceived enforcement probabilities are not directly observed.
- Strategic interaction also matters: one generator's exit changes rivals' expected future profits.
- A full dynamic game with all rival states would be computationally intractable.

4.2 What identifies perceived enforcement probabilities

- The key comparison is between:
 - generators subject to **MATS**, and
 - generators subject to **state air-toxics standards**.
- The identifying idea is difference-in-differences-like:
 - conditional on observables and predicted profits,
 - state-standard generators reveal behavior under much less policy uncertainty,
 - MATS generators reveal behavior under federal policy uncertainty.
- If MATS generators adopt or exit *more slowly* than comparable state-standard generators, the model interprets this as evidence of a lower perceived enforcement probability.
- The model allows MATS adoption costs to differ from state-standard adoption costs because state standards were often weaker.
- Useful reduced-form support: at the start of 2012, 8.6% of MATS generators newly reported retirement plans for 2012–15, versus only 1.4% of generators already under state standards.

4.3 What identifies adoption and exit costs

- The paper observes **two** dynamic actions:
 - adoption,
 - exit.
- This helps separate parameters:
 - the relative frequency of adoption vs. exit helps identify A and X ,

- observed annual profits help scale the dynamic choice problem in dollars.
- Because annual profits are estimated outside the long-run dynamic step, the scale parameter $1/\sigma$ is also identified in dollar terms.
- Within-year operating data identify ramping and O&M costs, so the long-run model is not trying to infer everything only from annual adopt / exit choices.

4.4 Why the application is credible, and limitations

- The profit surface includes US-state fixed effects, heat rate, capacity, and interactions with market states, which helps absorb persistent cross-state profit differences.
- The paper does *not* include year fixed effects in the dynamic step, because MATS timing is common across units and such fixed effects would be poorly identified and awkward for counterfactuals.
- The model does not separately identify annual fixed operating costs from exit costs, so the estimated exit cost should be interpreted as the present value of exit / remediation costs net of avoided future fixed operating costs.
- Main limitations:
 - no explicit coordination across generators of the same owner,
 - no endogenous coal entry,
 - non-IPP coal evolution is held fixed in counterfactuals,
 - credibility of ABOE depends on the three aggregate states capturing the key strategic forces.

4.5 Relation to the literature

- **Uncertainty literature.** The paper goes beyond measuring uncertainty by recovering firms' subjective enforcement probabilities and simulating timing-of-resolution counterfactuals.
- **Electricity-market structural IO.** It builds a dynamic oligopoly model of generator adoption and exit under uncertain environmental regulation.
- **Approximate equilibrium methods.** ABOE extends the literature on oblivious / approximate equilibria by letting *all* firms understand that their own actions move aggregate states, not just a subset of dominant firms.

5 Tables and figures

5.1 Table 1 and Table 2: sample and compliance behavior

Read:

- Table 1 shows that the two groups of generators are broadly comparable technologically, but not identical. MATS generators face higher coal fuel prices and therefore higher marginal fuel costs.

TABLE 1
GENERATOR DESCRIPTIVE STATISTICS BY REGULATORY REGIME

	US State Standards	MATS
Capacity (MW)	279.69 (213.78)	246.01 (286.83)
Heat rate (MMBtu/MWh)	10.51 (2.25)	11.99 (4.71)
Coal fuel price (\$/MMBtu)	1.77 (.17)	2.20 (.39)
Marginal fuel costs (\$/MWh)	18.48 (3.93)	27.29 (14.27)
Generators	93	217
Generator-years	841	1,950

SOURCE.—Authors' calculations are based on an annual analysis sample of IPP coal generators.

NOTE.—Standard deviations are shown in parentheses.

TABLE 2
COUNTS OF GENERATORS ADOPTING ABATEMENT TECHNOLOGY OR EXITING

Years to Enforcement	US State Standard			MATS		
	Adoptions	Exits	Share Complied	Adoptions	Exits	Share Complied
4	12	0	.34	2	23	.30
3	5	1	.51	1	18	.53
2	9	0	.77	2	7	.64
1	4	4	1.00	9	21	1.00
Total	30	5		14	69	

SOURCE.—Authors' calculations are based on an annual analysis sample of IPP coal generators.

- Table 2 is economically important: state-standard generators mostly comply by *adopting*, while MATS generators mostly comply by *exiting*.
- Together, the two tables explain why a simple raw comparison is not enough: the paper needs a structural model because the two groups face different market conditions and different regulatory regimes.
- Early adoption and exit before the deadline also show that firms are responding to idiosyncratic shocks and strategic considerations, not just mechanically to the final deadline.

5.2 Figure 1: adoption paths and implied beliefs

Read:

- The figure compares adoption shares by years to enforcement.
- Four years before enforcement, adoption rates under MATS and state standards look fairly similar.
- But three and two years before enforcement, MATS adoption is much lower than state-standard adoption.
- This is the key reduced-form evidence that firms perceived the probability of 2016 MATS enforcement to be well below one in the middle years.
- So the structural belief estimates are not coming out of nowhere; Figure 1 already points in that direction.

5.3 Table 3 and Figure 2: hourly operations and ramping costs

Read:

- Table 3 shows that coal units are not just steady baseload machines: they are off 31% of the time and at minimum generation 24% of the time.
- Figure 2 shows the ordering of ramping costs:
 - ramping from minimum to maximum is cheapest,
 - from off to minimum is more costly,

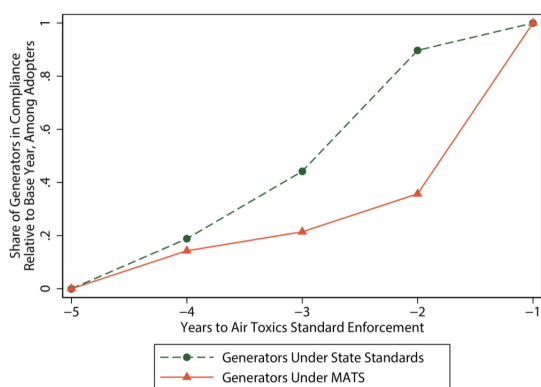


FIG. 1.—Share of generators that have adopted air toxics abatement technology among generators subject to US state air toxics standards, relative to the total number of generators that had not adopted abatement technology 5 years before air toxics standard enforcement but that eventually adopted. Calculations are based on annual analysis sample.

- from off to maximum is most costly.
- This pattern says start-up costs matter a lot.
- Ramping costs rise strongly with generator capacity.
- The paper also estimates average O&M costs of about \$15.55/MWh, close to engineering benchmarks.
- These within-year operating costs are crucial because they determine the annual profit object feeding into the dynamic exit / adoption model.

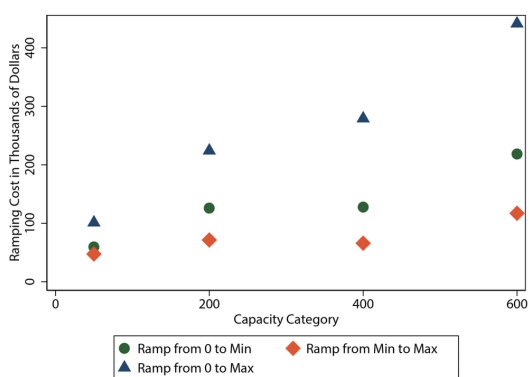


FIG. 2.—Variation in ramping cost estimates with capacity. Ramping costs are ratios of estimated ramping coefficient to operating revenue coefficient from separate regressions by capacity bin. Symbols are placed at the midpoints of the capacity bins. The figure omits estimates for generators larger than 700 MW.

TABLE 3
DESCRIPTIVE STATISTICS OF HOURLY GENERATION

Statistic	Value
Share of hours at maximum generation	.45 (.50)
Share of hours at minimum generation	.24 (.43)
Share of hours at zero generation	.31 (.46)
Generation (MWh)	177.02 (250.96)
Electricity price (\$/MWh)	39.38 (25.14)
US state electricity demand (GWh)	13.47 (5.37)
Daily heating degrees	13.91 (14.91)
Daily cooling degrees	2.83 (4.82)
Observations	25,194,607

SOURCE.—Authors' calculations are based on hourly generation data of IPP coal generators.

NOTE.—Standard deviations are shown in parentheses.

TABLE 4
PROFIT AND POLLUTION SURFACE REGRESSION RESULTS

	Annual Profit (\$ in Millions) (1)	Log Annual Pollution (Pounds of SO ₂) (2)
Compliant		-.758*** (.063)
US state coal share	-21.984*** (6.266)	2.155*** (.275)
US state coal share ²	10.777*** (1.953)	-.129* (.076)
Gas-to-coal fuel price ratio	5.181*** (.873)	.467* (.250)
Gas-to-coal fuel price ratio ²	-.625*** (.097)	.145** (.070)
Coal share × fuel price ratio	-1.141*** (.327)	.127** (.055)
Heat rate (MMBtu/MW)	-.313*** (.113)	.582** (.234)
Capacity (MW)	-.035*** (.005)	-.389 (.244)
Capacity ² (MW ²)	-2.3e-05*** (4.0e-06)	.112*** (.021)
Capacity × US state coal share	-.007*** (.002)	-.032 (.026)
Capacity × fuel price ratio	.048*** (.001)	-.005 (.039)
US state fixed effects	Yes	Yes
Observations	2,899	2,718
R ²	.7791	.7001

NOTE.—Regression of calculated profits and log pollution from observed data on dynamic model states. The regressions exclude generator-years with zero production. The pollution regression uses logs of the independent variables other than compliance and excludes generator-years with zero pollution. Standard errors are shown in parentheses.

* Significant at 10%.

** Significant at 5%.

*** Significant at 1%.

5.4 Table 4 and Figure 3: profit and pollution surfaces, and industry evolution

Read:

- The estimated profit surface has $R^2 = 0.7791$, and the pollution surface has $R^2 = 0.7001$. So the reduced-form surfaces fit reasonably well.
- Higher heat rate lowers profits, as expected.
- Higher coal share tends to lower profits, which matches oligopoly logic: more coal capacity means tougher competition.
- A higher gas-to-coal fuel price ratio tends to raise coal profits, because coal becomes relatively more competitive.
- Compliance sharply reduces pollution in the pollution regression.
- Calculated annual profits are often low: the 5th percentile is about $-\$7.9$ million, the 95th percentile about $\$99$ million, and only 62% of profits are above zero.
- Figure 3 shows the big industry story:
 - gas prices fall after fracking,
 - coal generator profits fall,
 - coal capacity exits with a lag,
 - predicted mean profits track calculated profits well over time.

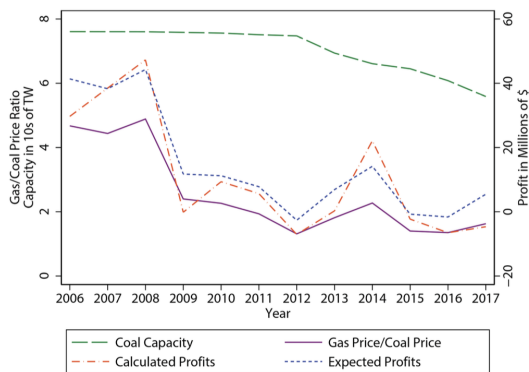


FIG. 3.—Variation in coal capacity, mean profits, and fuel prices over time. Authors' calculations are based on the analysis sample of IPP coal generators. Calculated profits use our estimated ramping and O&M costs and observed generation choices. Expected mean profits are predicted for generator-years in our sample using estimated profit surface regression parameters.

TABLE 5
STRUCTURAL PARAMETER RESULTS

	Base Specification (1)	Same Adoption Cost State vs. MATS (2)
Predicted enforcement probabilities:		
Probability 2012	1.000*** (.082)	.989*** (.058)
Probability 2013	.738*** (.150)	.623*** (.128)
Probability 2014	.428** (.179)	.297** (.130)
Probability 2015	1.000*** (.072)	.999*** (.072)
Generator costs:		
Adoption cost (\$ in millions)	312.2*** (113.1)	694.5*** (90.7)
Extra MATS adoption cost (\$ in millions)	570.6*** (72.9)	
Exit scrap value (\$ in millions)	−326.5*** (75.5)	−313.8*** (68.6)
1/σ (\$ in millions)	40.3*** (4.4)	41.2*** (5.4)
Simulated log likelihood	−616.97	−624.76

NOTE.—Structural parameter estimates from nested fixed-point estimation. Standard errors calculated via a parametric bootstrap are shown in parentheses.

** Significant at 5%.

*** Significant at 1%.

5.5 Table 5: structural estimates of beliefs and costs

Read:

- This is the paper's main structural table.
- Estimated perceived enforcement probabilities:
 - 2012: 1.000,

- 2013: 0.738,
 - 2014: 0.428,
 - 2015: 1.000.
- The 2014 collapse is the headline result and lines up with the period when the Supreme Court agreed to hear *Michigan v. EPA*.
 - Adoption cost under state standards is about \$312 million.
 - MATS compliance requires an *extra* \$571 million on top of that, so implied MATS adoption cost is roughly \$883 million.
 - Exit scrap value is about –\$326.5 million, meaning exiting is very costly.
 - Since mean generator profit in the sample is only about \$17 million, these are economically huge costs.
 - The alternative specification with equal adoption costs changes levels somewhat, but the time pattern of beliefs is very similar.

5.6 Table 6: model fit and counterfactuals

Read:

- The model fits adoption fairly well: it predicts 14.5 adopters versus 14 in the data.
- Baseline estimated model (2012–2041 discounted totals):
 - adoption costs: \$11.77B,
 - exit costs: \$30.47B,
 - total profits: \$64.42B,
 - pollution: 785.72 million pounds of SO₂.
- The clean timing-of-resolution comparison is column 3 vs. column 4:
 - same mean enforcement probability = 0.7913,
 - only timing differs.
- Resolving uncertainty in 2012 instead of 2016:
 - raises profits by \$1.39B,
 - lowers adoption costs by about \$0.66B,
 - lowers exit costs by about \$0.68B,
 - increases the 2016 generator count from 127.2 to 131.4,
 - raises pollution from 793.75 to 846.26 million pounds.
- That pollution increase is about 52.5 million pounds, with damages valued at \$652M–\$1.776B.
- The no-exit-cost counterfactual sharply increases exit:
 - 2016 generators fall to 106.4,

- pollution falls to 575.10 million pounds,
- profits jump to \$102.79B.
- But that counterfactual requires the government, not firms, to absorb exit costs.

TABLE 6
MODEL FIT AND COUNTERFACTUAL RESULTS

	Data (1)	Estimated Model (2)	Same Mean Probability: .7913 (3)	Uncertainty Resolved in 2012 with Mean Probability (4)	No Exit Cost (5)
Adoption costs (\$ in billions)		11.77	11.28	10.62	7.93
Exit costs (\$ in billions)		30.47	30.27	29.59	0.00
Total profits (\$ in billions)		64.42	65.96	67.35	102.79
Pollution (millions of lbs of SO ₂)		785.72	793.75	846.26	575.10
Number of generators:					
2012	183	183.0	183.0	183.0	183.0
2013	160	169.8	170.4	170.6	163.4
2014	142	158.1	158.3	158.8	146.4
2015	135	148.7	146.7	147.5	132.7
2016	114	125.9	127.2	131.4	106.4
Count adopting	14	14.5	13.8	12.98	9.7

NOTE.—Column 1 reports observed exit and adoption decisions in the data. Column 2 reports predicted outcomes at model estimates. Column 3 replaces the estimated probabilities of 2016 MATS enforcement with the mean estimated probability across years. Column 4 calculates the expected outcomes with uncertainty completely resolved in 2012, with the mean estimated probability across years. Column 5 sets exit costs to zero. The first four rows of results report the total discounted profits or costs from 2012 through 2041.

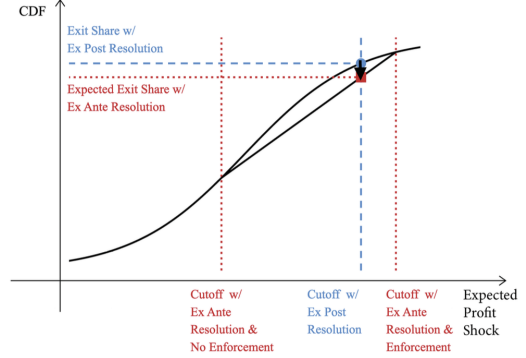


FIG. 4.—Conceptual CDF for a single generator's expected profits in a given year. Cutoffs indicate the minimum expected profit necessary for the generator to remain in the market. Thus, the CDF represents the share of expected profit draws that are below a given cutoff and lead to generator exit.

5.7 Figure 4: why early resolution can increase pollution

Read:

- Figure 4 is the conceptual mechanism figure.
- The blue dashed line is the ex post uncertainty cutoff for staying in the market.
- The red dotted lines are the ex ante cutoffs with and without enforcement.
- When many generators are already close to the exit margin, the probability reduction from learning “no enforcement” can outweigh the probability increase from learning “enforcement.”
- In that case, earlier uncertainty resolution *reduces expected exit*.
- Since coal generation creates negative externalities, less exit means more pollution.
- This figure is the best way to understand why “less uncertainty” can improve firm profits while worsening environmental outcomes.

6 Short summary

- **Method.** The paper estimates a dynamic oligopoly model of technology adoption and exit under policy uncertainty and introduces ABOE to make equilibrium computation tractable.
- **Identification.** It compares MATS generators with generators under state air-toxics standards, using the latter as a relatively low-uncertainty benchmark to recover subjective enforcement probabilities.
- **Application.** In US coal electricity markets, compliance with MATS mainly happened through exit rather than adoption, and generators faced very large adoption and exit costs.

- **Main result.** Policy uncertainty around MATS was substantial, peaking in 2014. Resolving that uncertainty earlier would have increased generator profits, but also increased pollution because more coal generators would have stayed in the market.

7 Conclusion

7.1 Core conclusion

- The paper shows that policy uncertainty can be measured structurally from firms' dynamic choices.
- It also shows that large dynamic games can be estimated in a strategically meaningful way without tracking the full micro-state of every rival.
- In this application, earlier resolution of MATS uncertainty is *not* a clear welfare improvement once pollution externalities are included.

7.2 Economic intuition

- Earlier resolution lets firms tailor exit and adoption decisions to the realized policy state rather than to an expectation, so private profits rise.
- But when many firms are close to the exit margin, revealing that regulation might not bind pulls many of them back into staying.
- With a negative externality like pollution, that means higher emissions.
- So the sign of earlier uncertainty resolution depends on the curvature of the profit distribution and how many firms are near the irreversible-decision margin.

7.3 Takeaway

This paper is important both methodologically and substantively. Methodologically, it gives a practical way to estimate a large dynamic oligopoly with policy uncertainty and strategic state effects. Substantively, it shows that faster policy resolution can help firms while hurting environmental goals. In the coal-electricity setting, the timing of uncertainty resolution is itself a major policy object, not just a side issue.